

Biophysics (Bio215)

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More hands-on

- Protein Structure prediction (modeller)

10 Feb (Tuesday)

Protein structure prediction: The holy grail of bioinformatics



Quick recap...

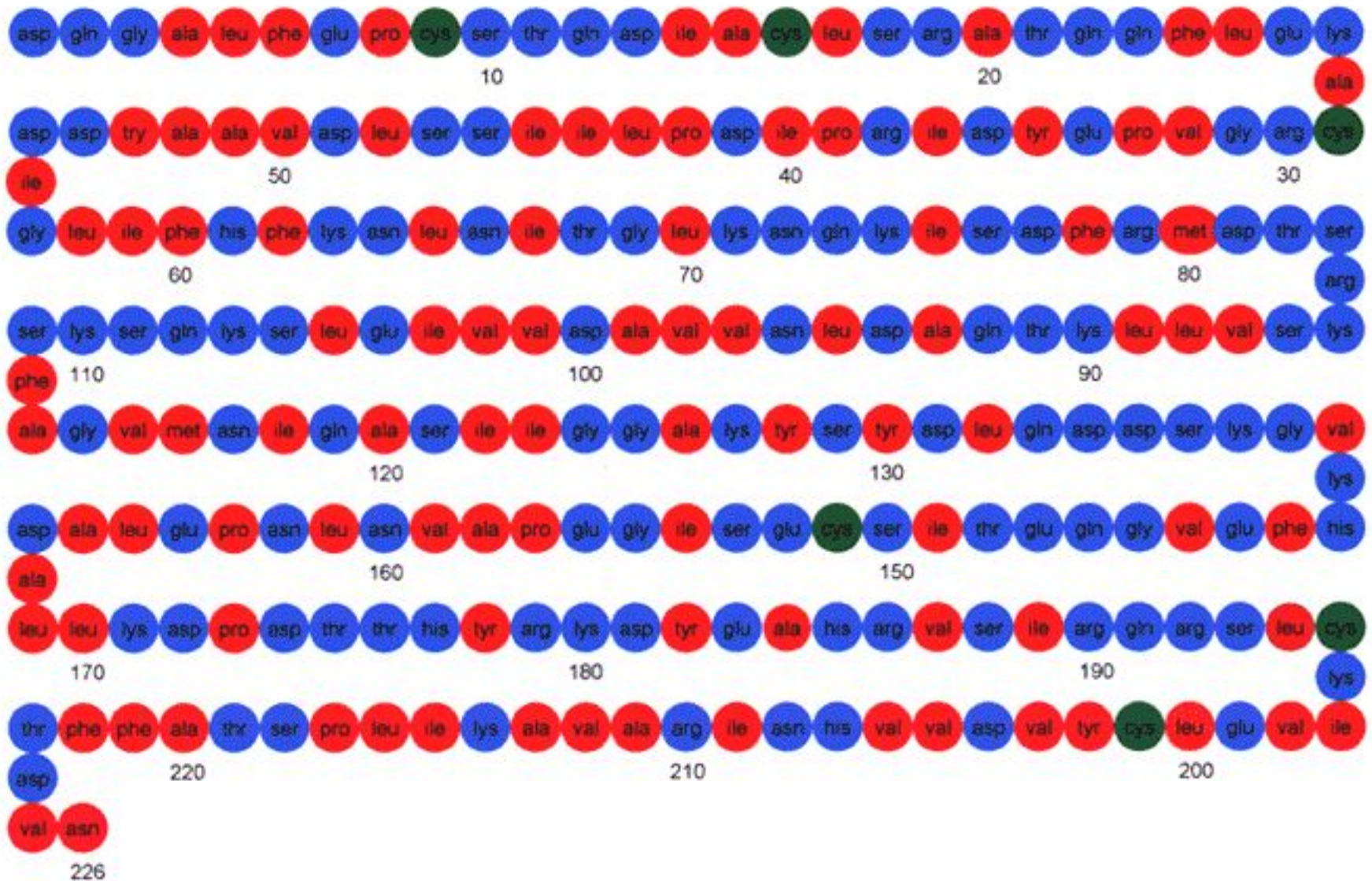
Proteins: Four levels of structural organization:

Primary structure

Secondary structure

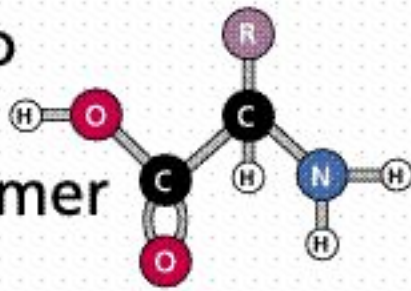
Tertiary structure

Quaternary structure

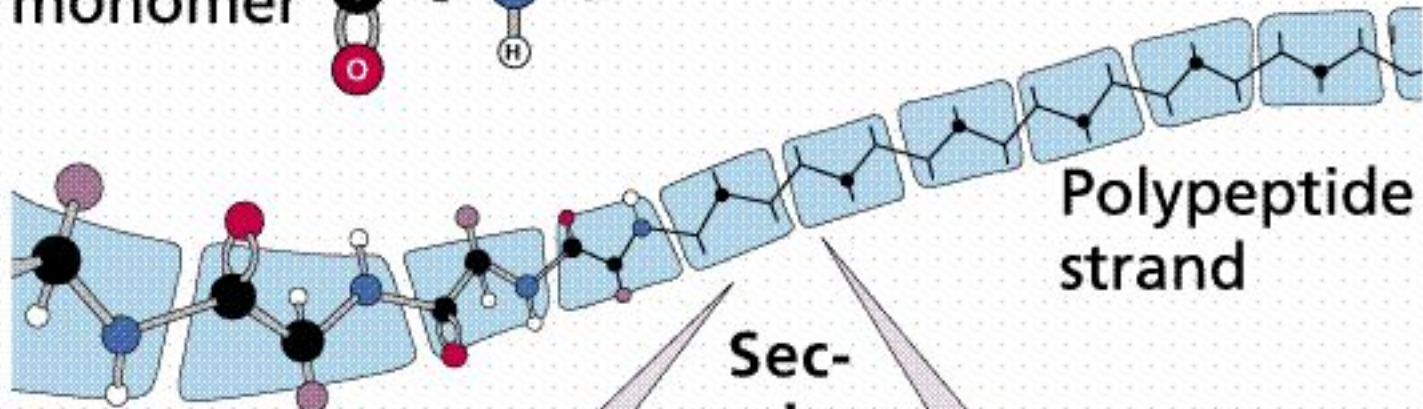


Primary structure = the linear amino acid sequence

Amino
acid
monomer



Primary Structure



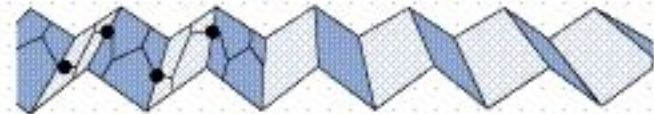
Polypeptide
strand

Sec-
ondary
Structure

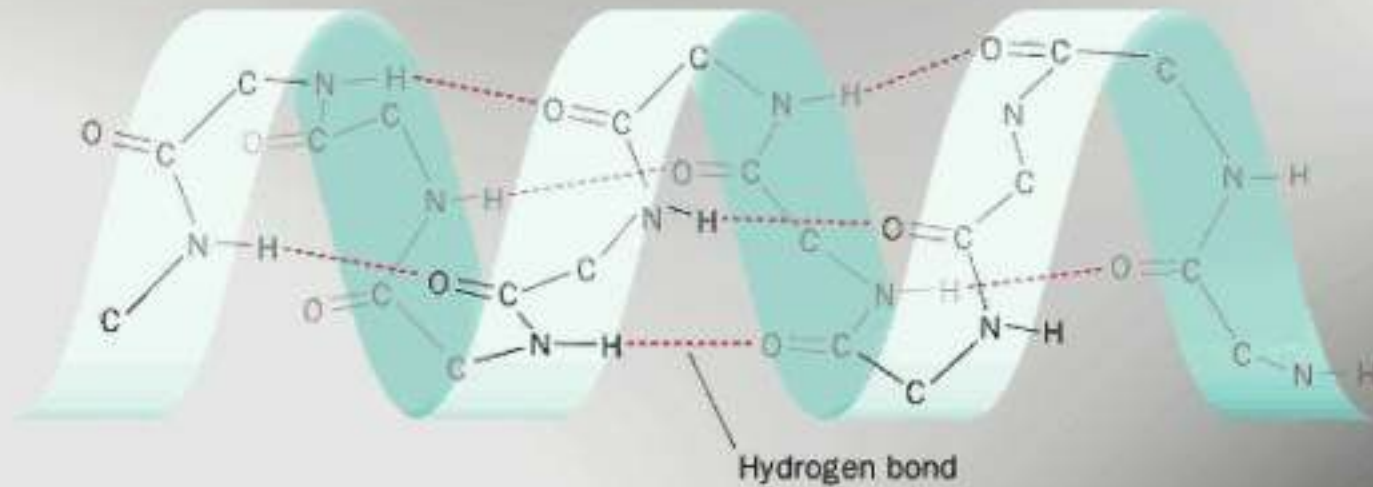
α Helix



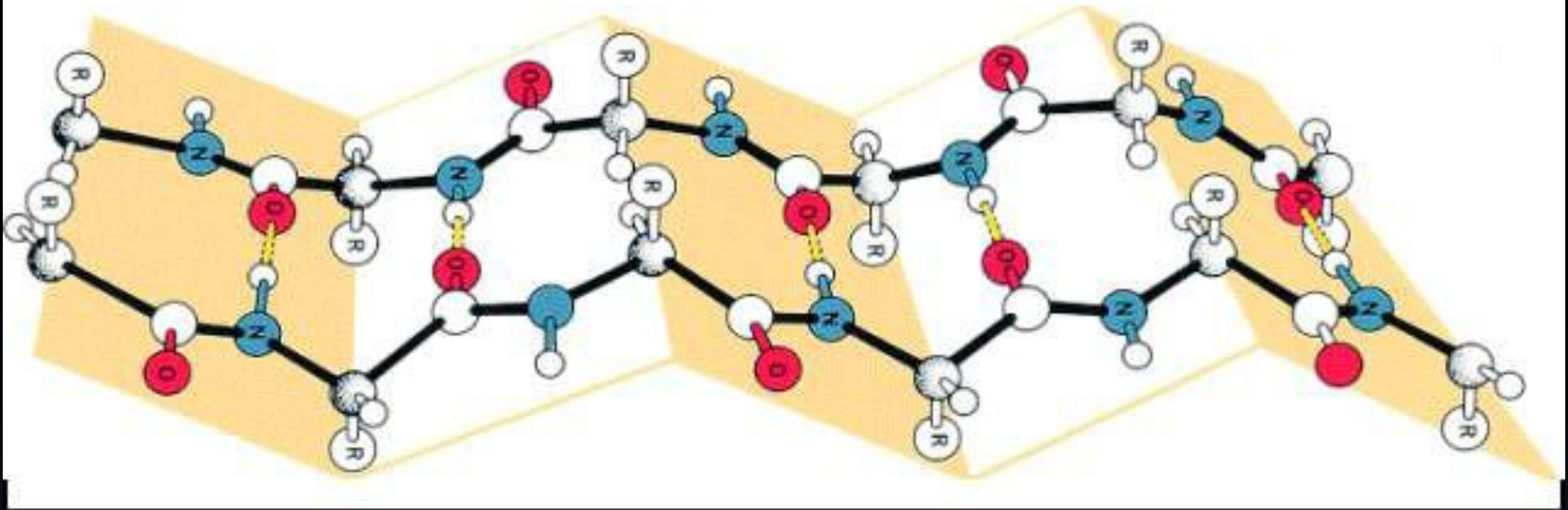
β -Pleated
sheet



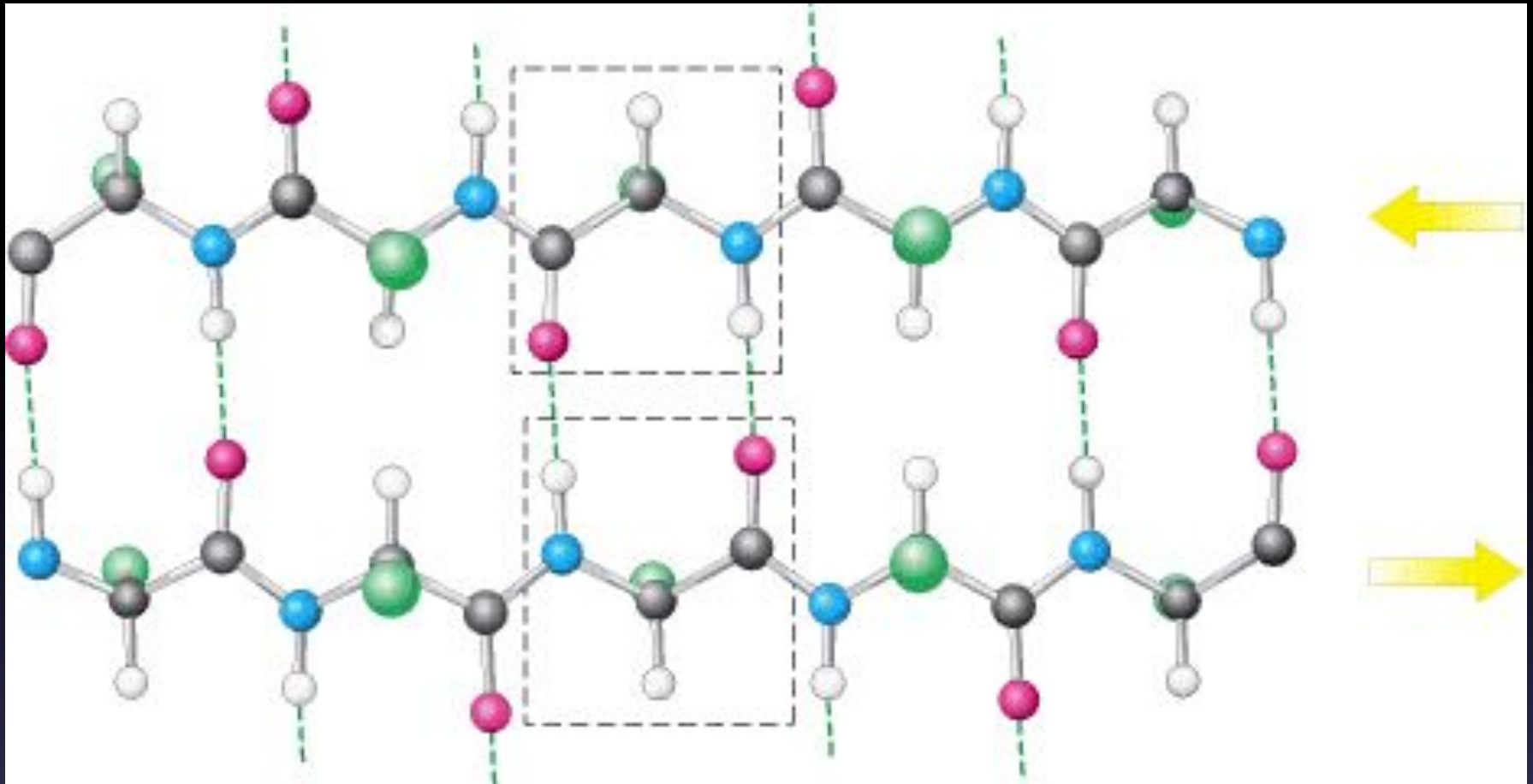
Secondary structure = spatial arrangement of amino-acid residues that are adjacent in the primary structure



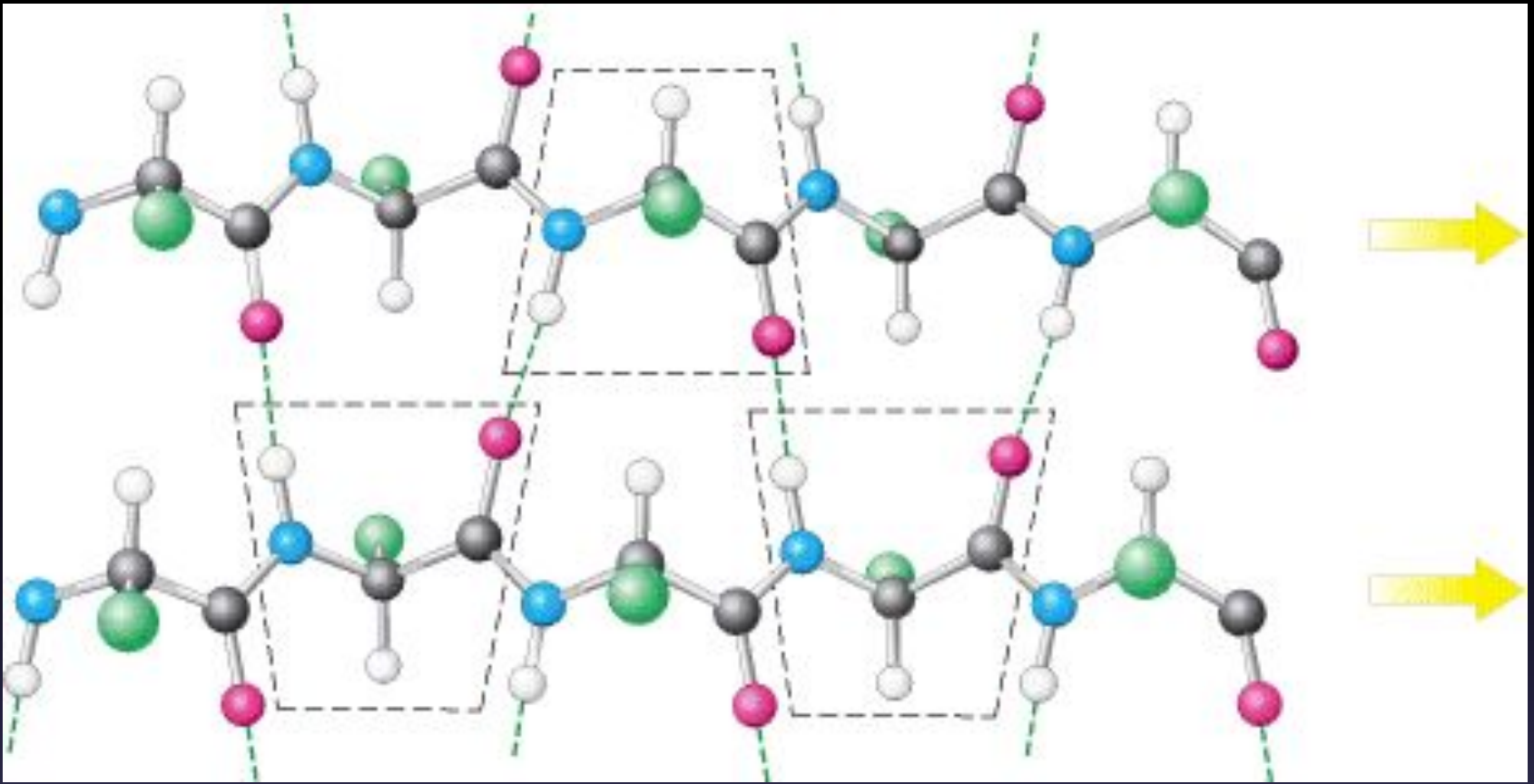
α helix = A helical structure, whose chain coils tightly as a right-handed screw with all the side chains sticking outward in a helical array. The tight structure of the α helix is stabilized by same-strand hydrogen bonds between $-\text{NH}$ groups and $-\text{CO}$ groups spaced at four amino-acid residue intervals.



The β -pleated sheet is made of loosely coiled β strands are stabilized by hydrogen bonds between -NH and -CO groups from adjacent strands.

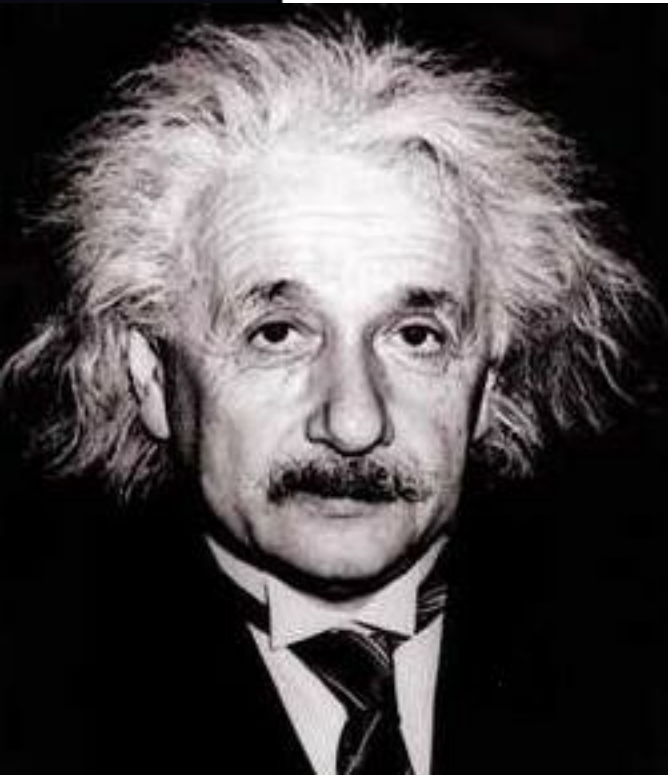


An antiparallel β sheet. Adjacent β strands run in opposite directions. Hydrogen bonds between NH and CO groups connect each amino acid to a single amino acid on an adjacent strand, stabilizing the structure.



A parallel β sheet. Adjacent β strands run in the same direction. Hydrogen bonds connect each amino acid on one strand with two different amino acids on the adjacent strand.

Keratin:



α -Helix



Coiled coil of two α -helices



Protofilament (pair of coiled coils)



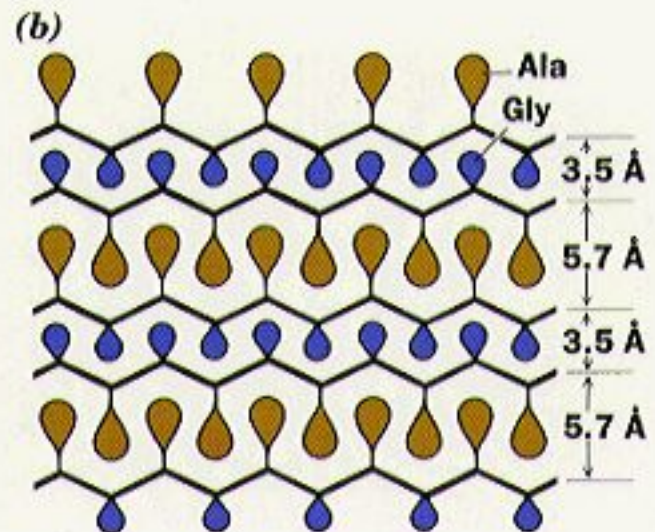
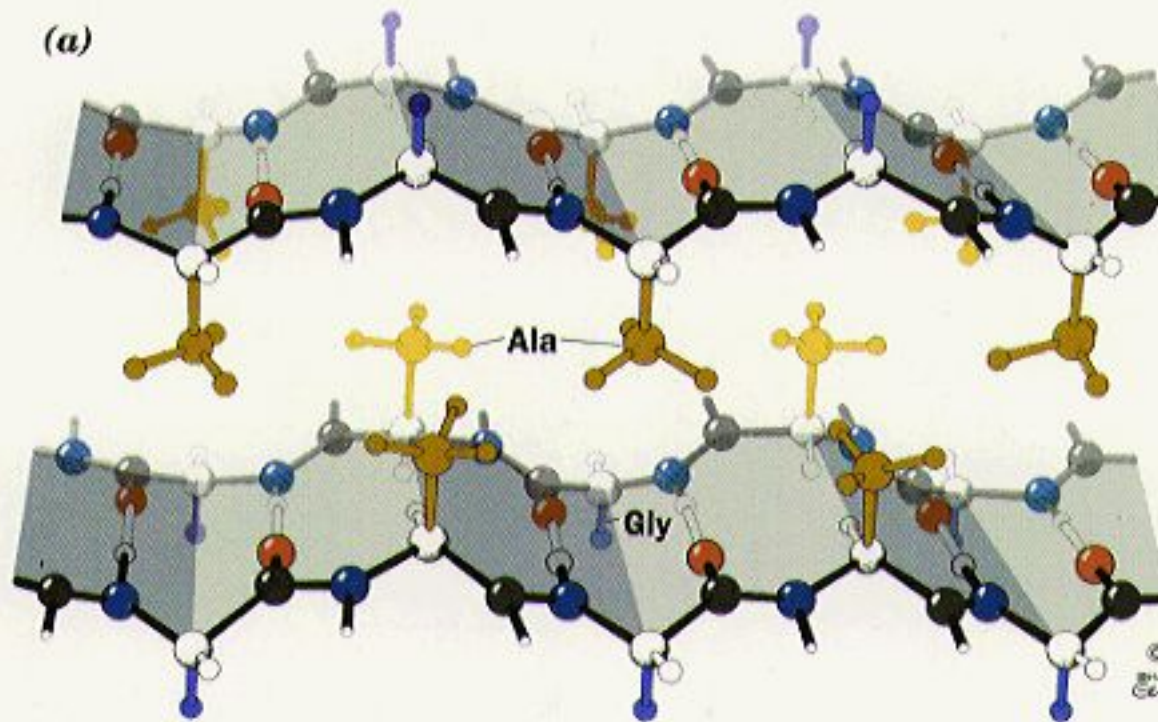
Filament (four right-hand twisted protofibrils)



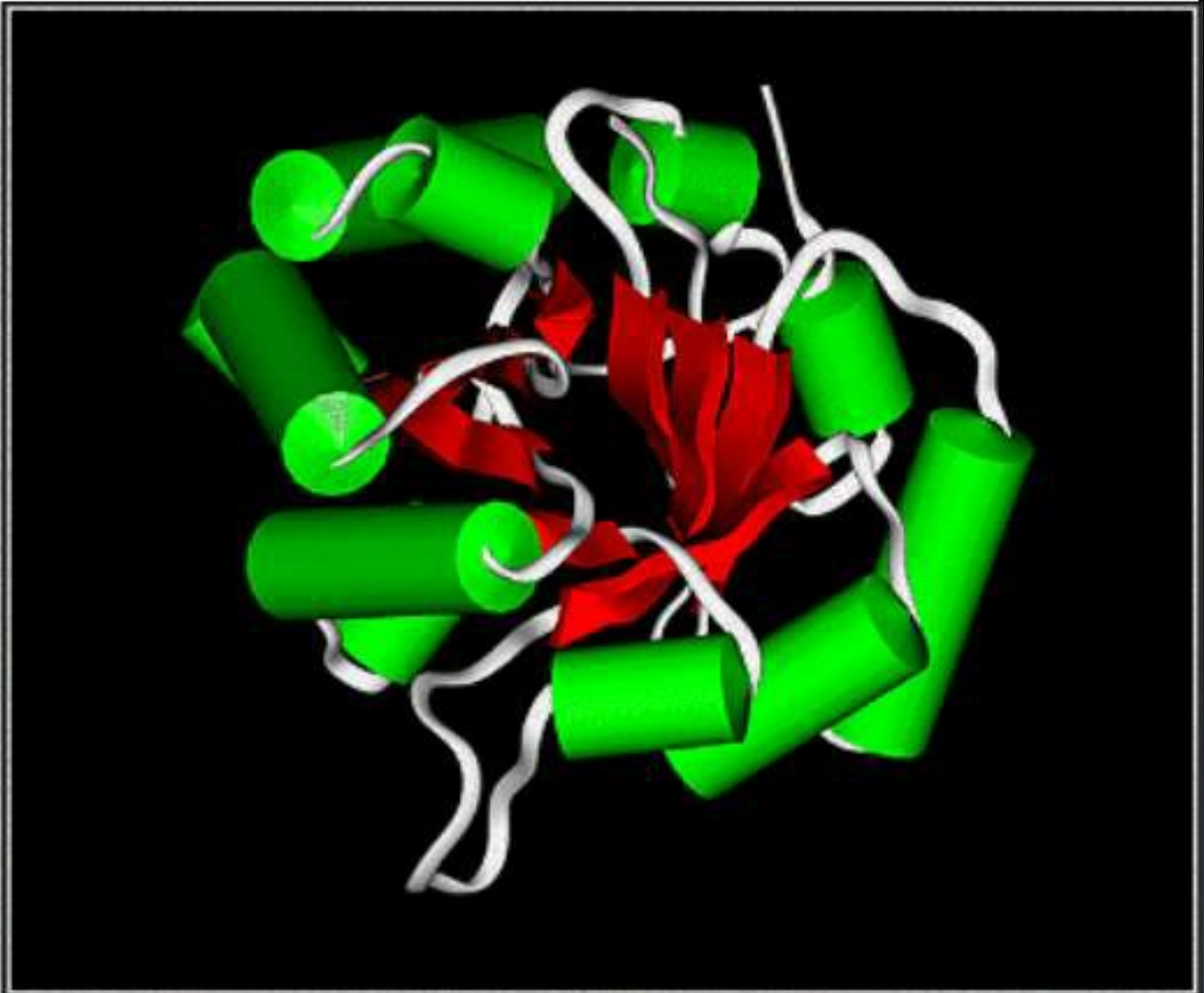


Silk

fibroin



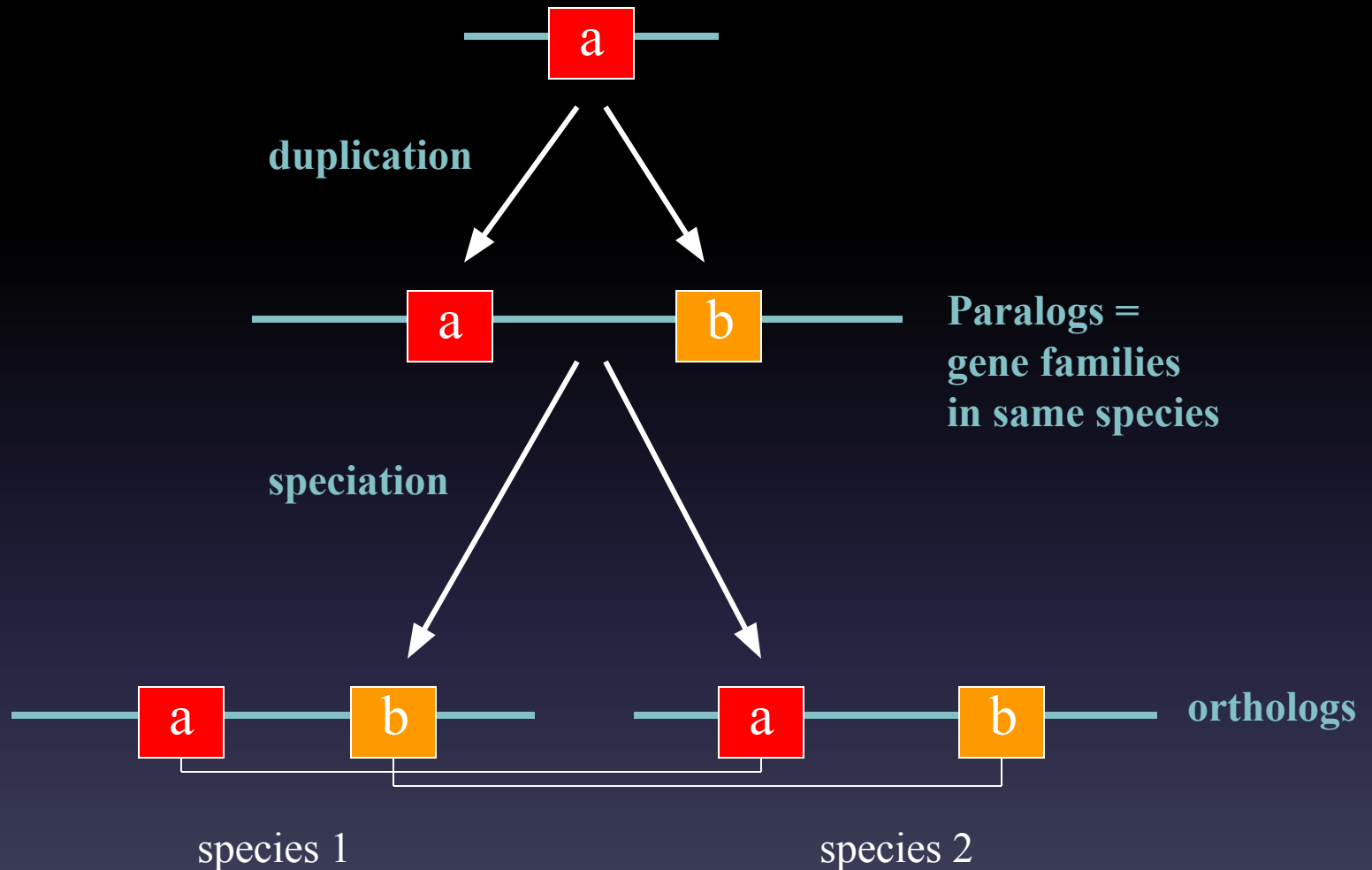
Tertiary
structure

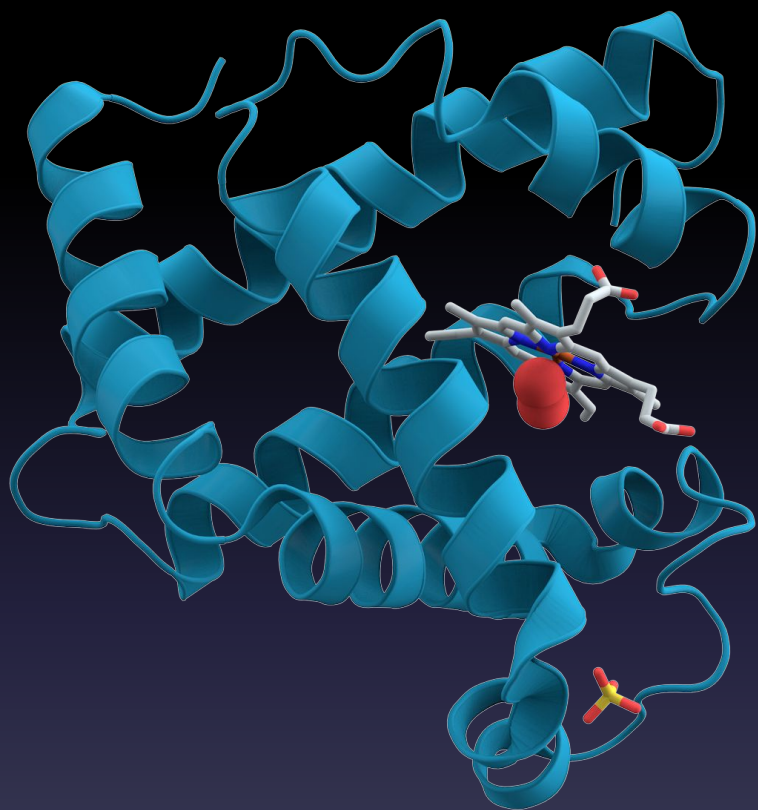


Structure of triosephosphate isomerase.

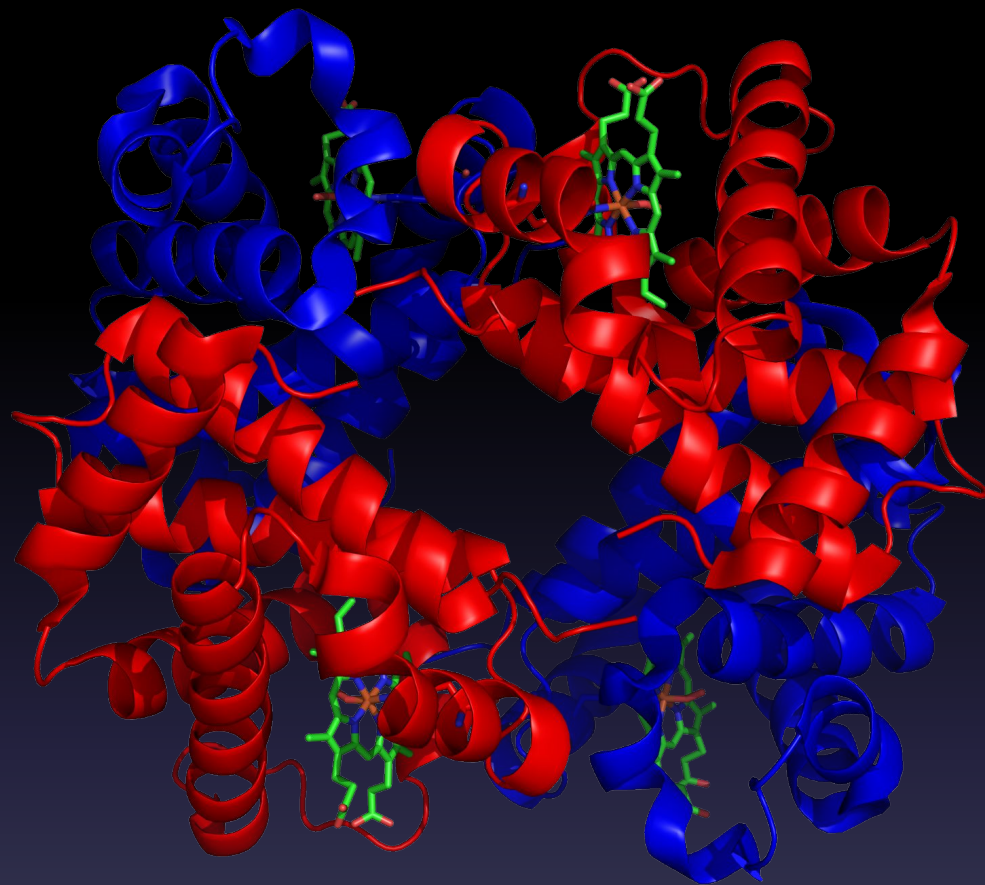
Tertiary structure = three-dimensional structure of protein

Homology: Paralogs and orthologs





Myoglobin

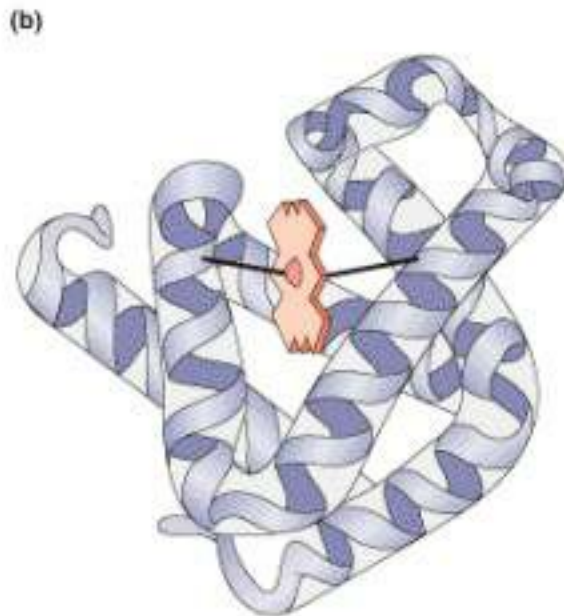


Hemoglobin

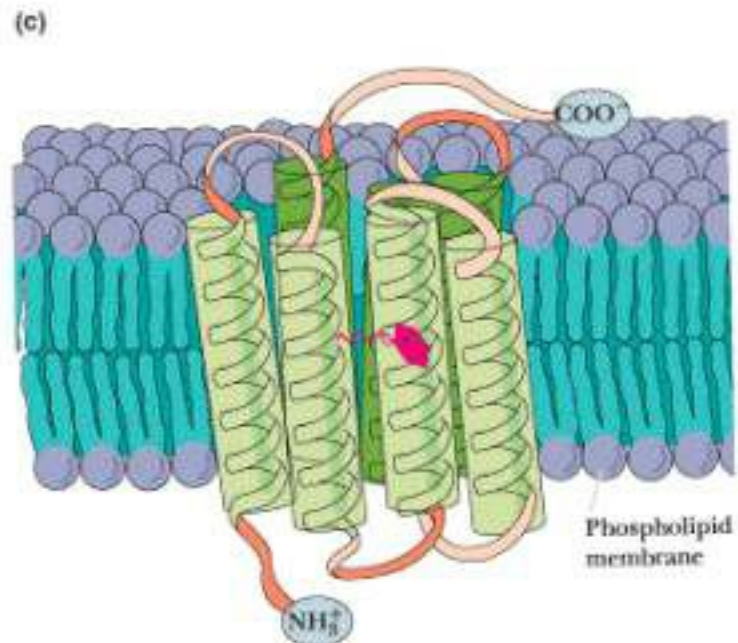
Fibrous, globular, and membrane proteins:



Collagen, a fibrous protein



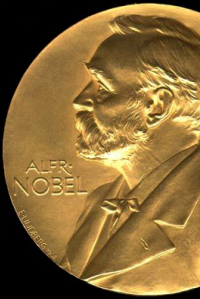
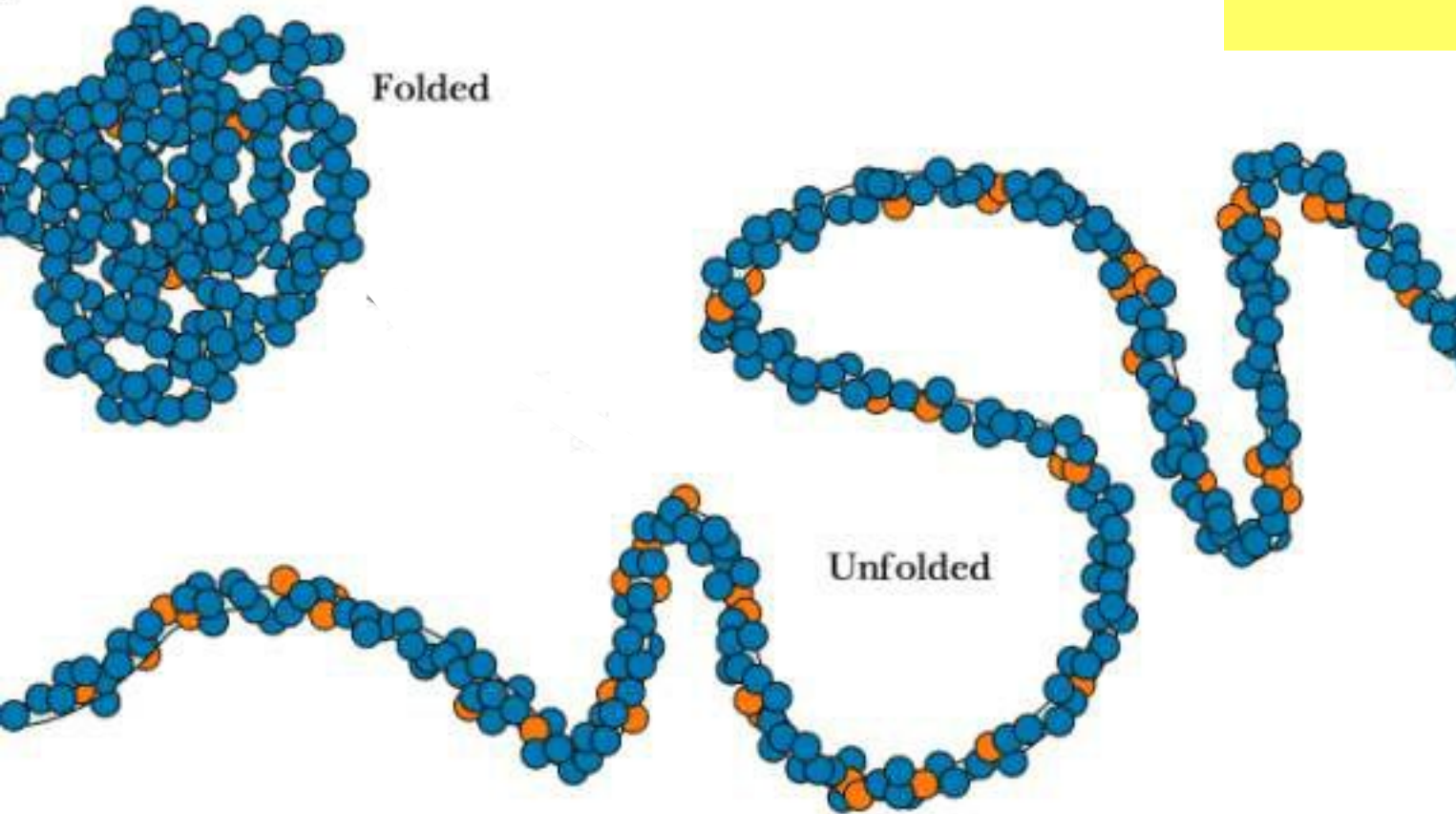
Myoglobin, a globular protein



Bacteriorhodopsin

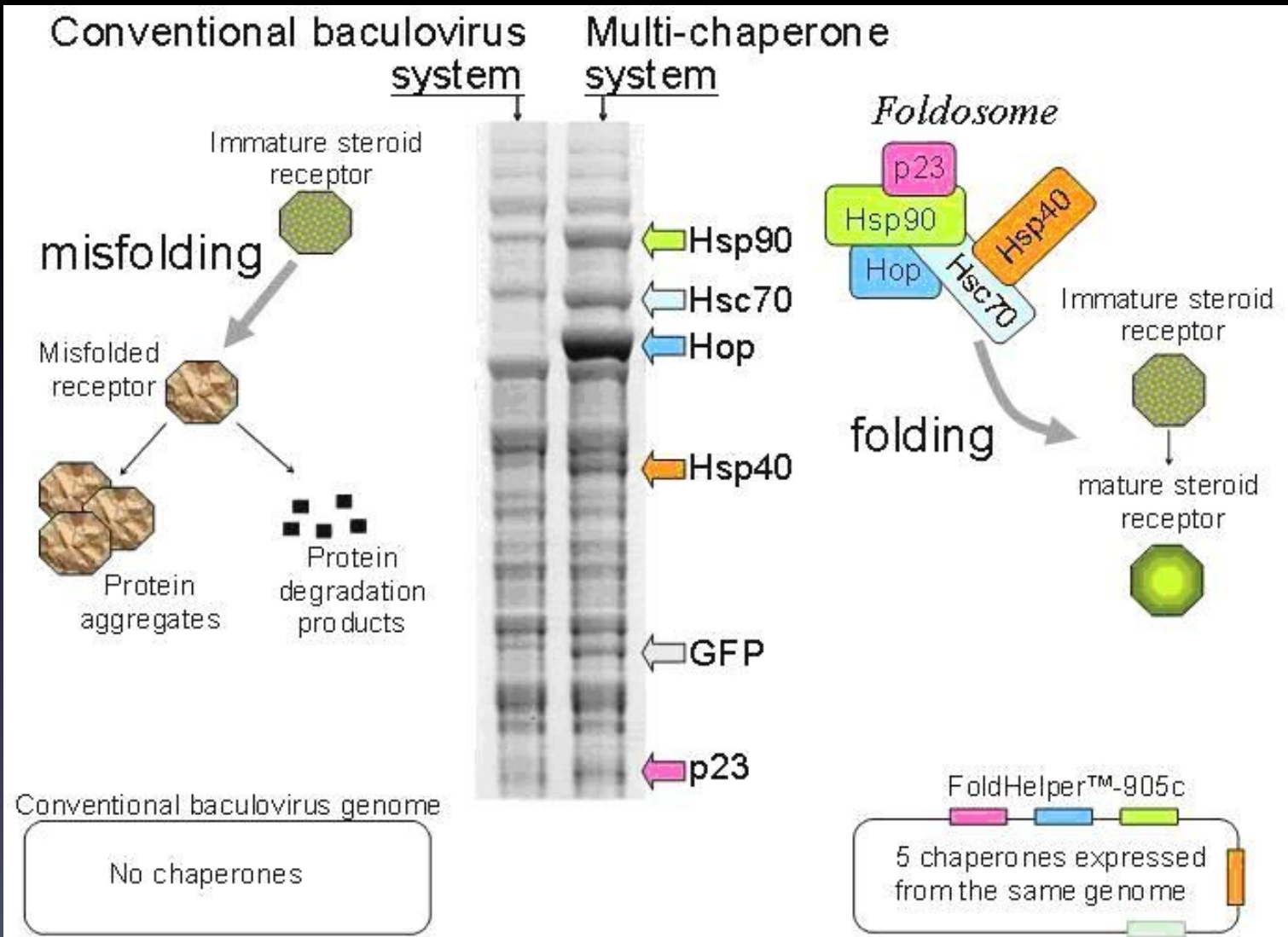
Folded vs. unfolded protein:

Christian B. Anfinsen
1916-1995

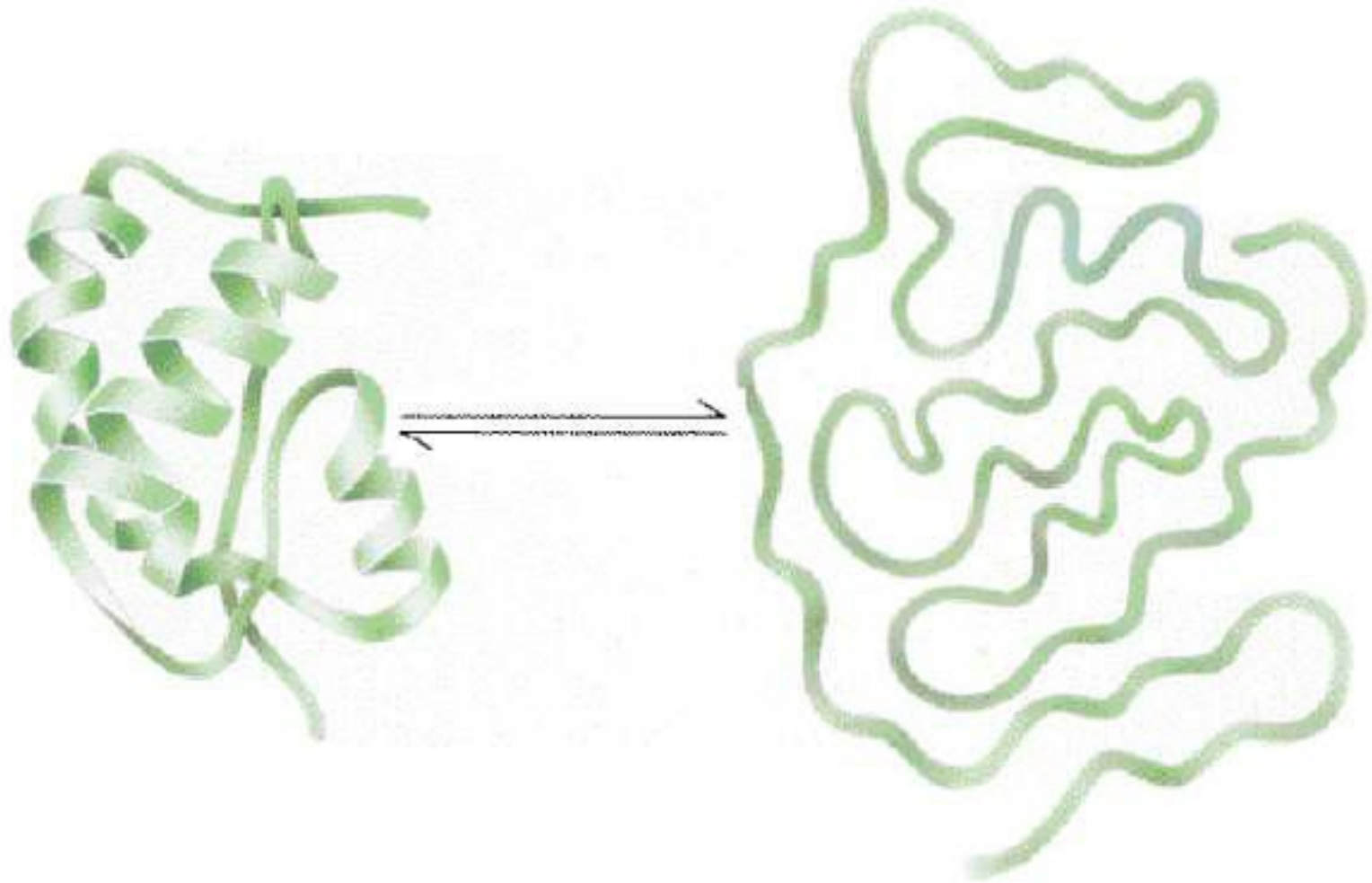


Sela M, White FH, & Anfinsen CB. **1959**. The reductive cleavage of disulfide bonds and its application to problems of protein structure. *Biochim. Biophys. Acta.* 31:417-426.

Not all proteins fold independently. Chaperones.



Native and denatured forms of a protein



Hair protein whose shape is fixed by disulfide bonds



STEP 1
Hair-releasing agent
Reduction



The protein now is not fixed by the disulfide bonds.

STEP 2
Hair is rolled onto curlers.



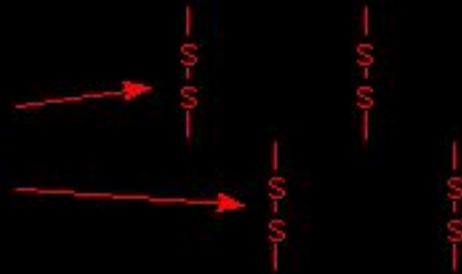
Different S-H bonds are now next to one another

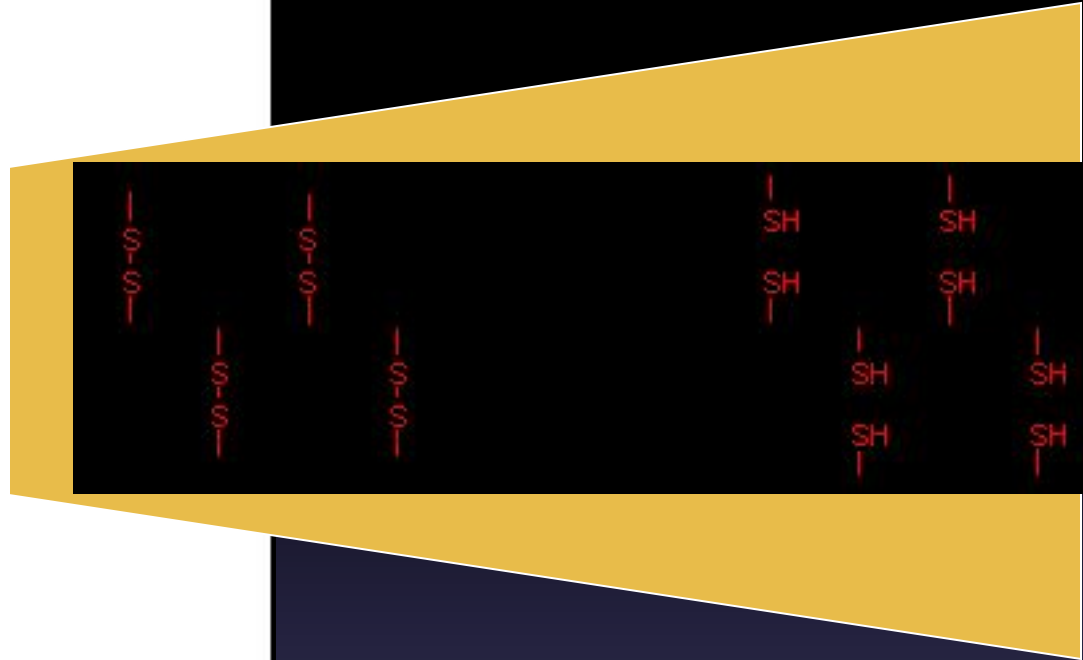
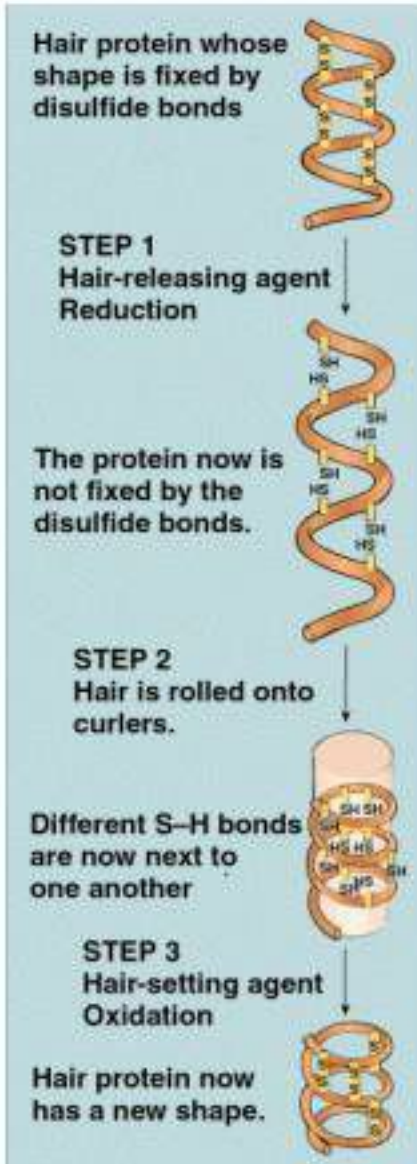
STEP 3
Hair-setting agent
Oxidation

Hair protein now has a new shape.

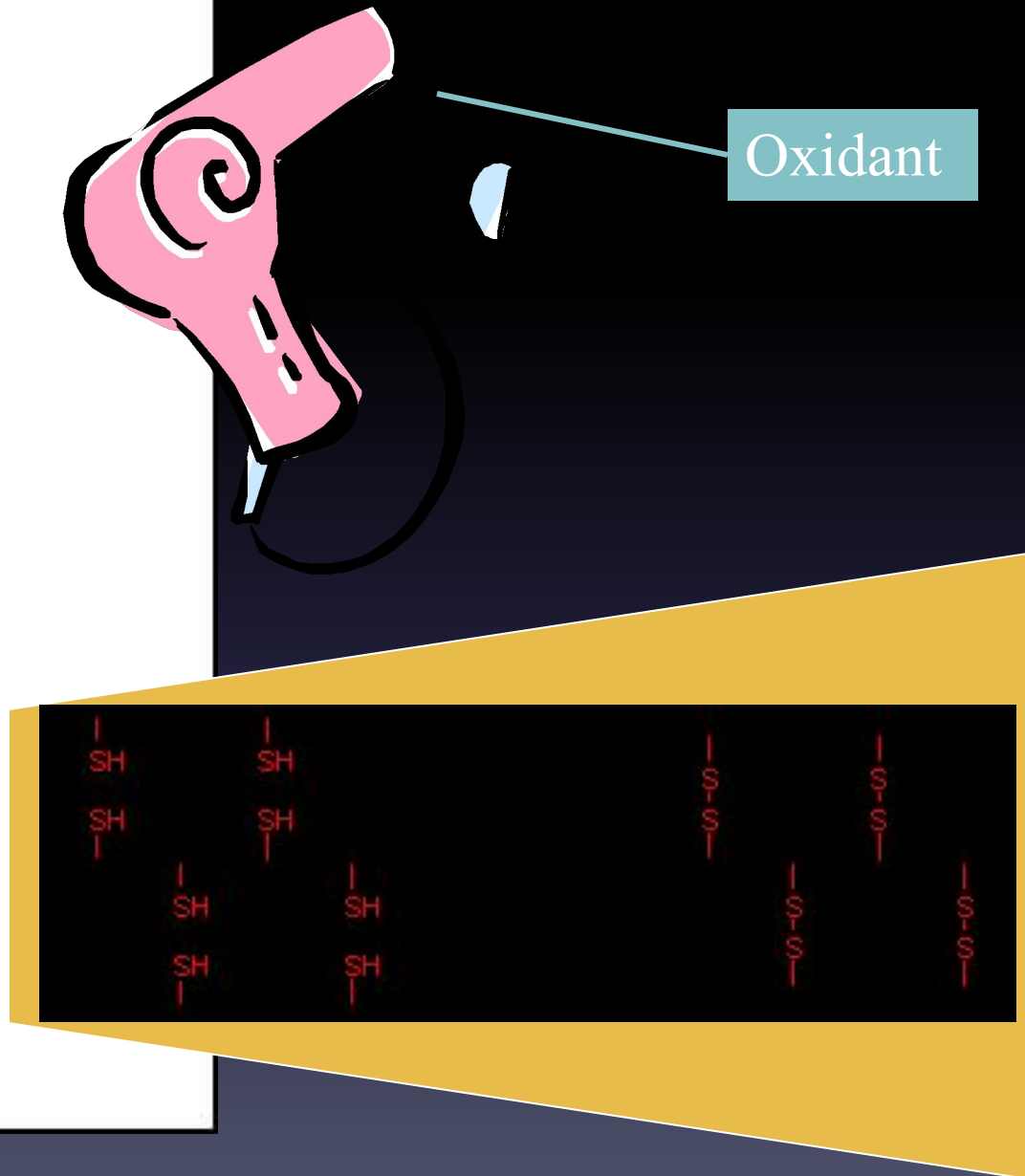
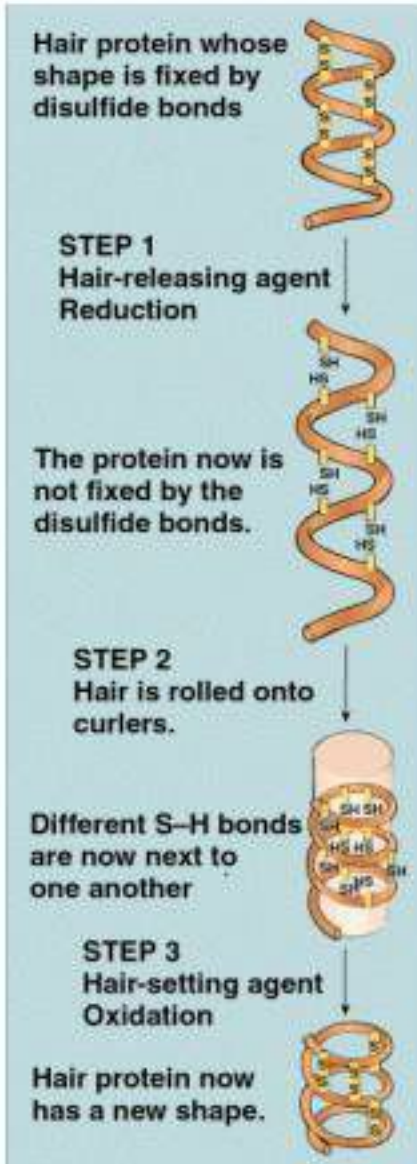


disulfide
cross-links





Reducing agents:
Ammonium thioglycolate (alkaline) pH 9.0-10
Glycerylmonothioglycolate (acid) pH 6.5-8.2





Protein Structure Classification

Class(C) secondary structure content – mainly alpha, mainly beta, alpha/beta, few secondary structures (type)

Architecture(A) gross arrangement of sec. structure elements (type and number of SS elements)

Topology(T) shape and connectivity of SS (type, number and order of SS elements)

Homologous superfamily (H)

Main classification levels



Class 1: Mainly Alpha



Class 2: Mainly Beta



Class 3: Mixed Alpha-Beta



Class 4: Few Secondary Structures

Roll	
Super Roll	
Barrel	
2-Layer Sandwich	
3-Layer(aba) Sandwich	
3-Layer(bba) Sandwich	
3-layer(bab) Sandwich	

Fold families

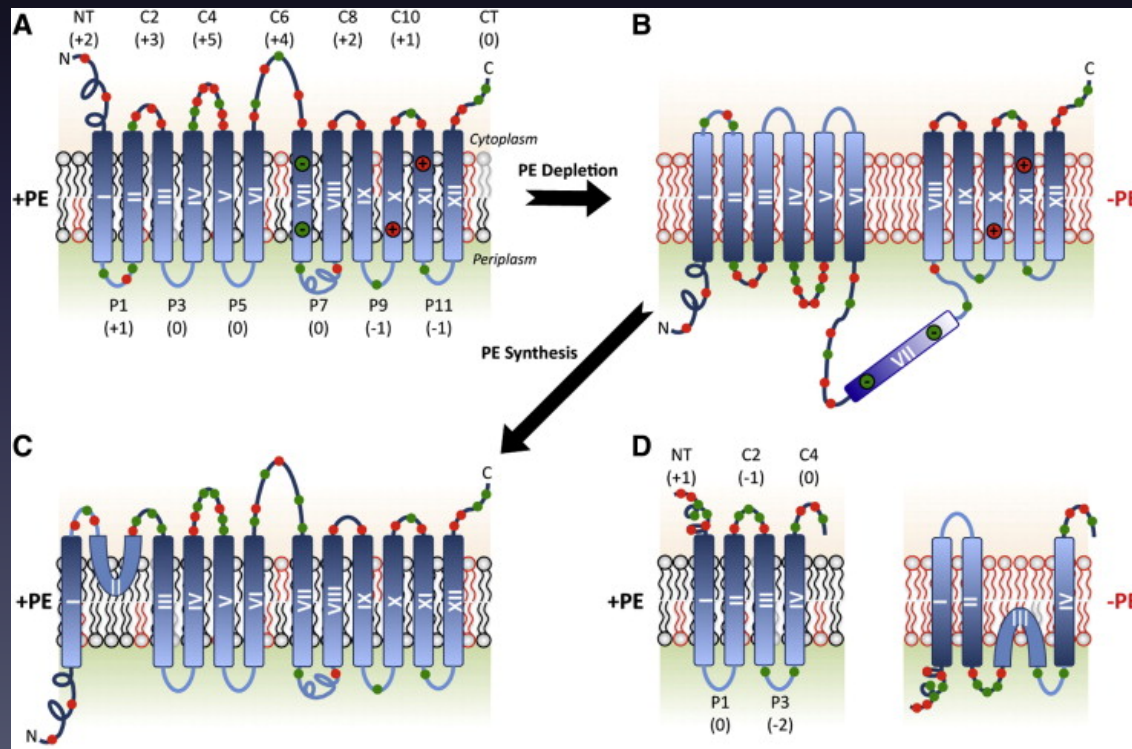
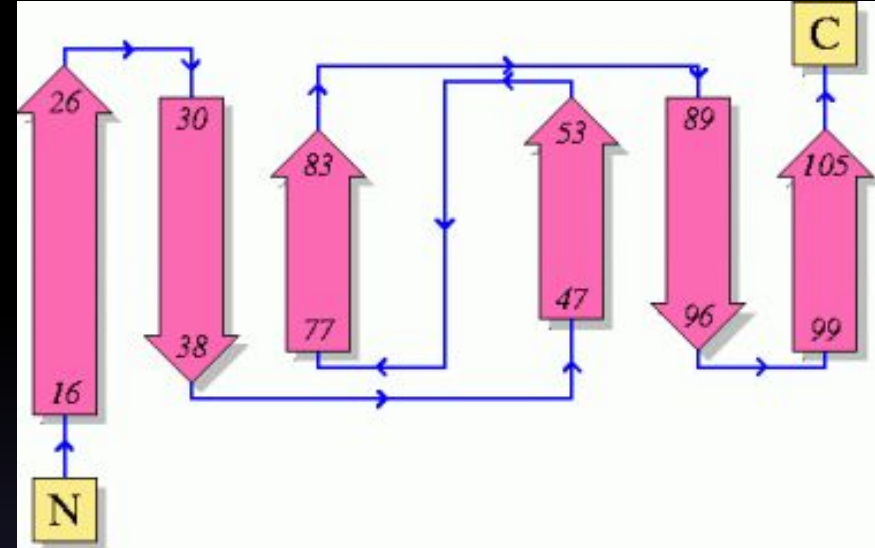
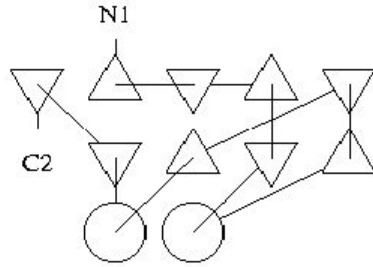
Homologous domains,
share common ancestor

Class

Architecture

Topology

H-level



Main classification levels



Class 1: Mainly Alpha



Class 2: Mainly Beta



Class 3: Mixed Alpha-Beta



Class 4: Few Secondary Structures

Roll	
Super Roll	
Barrel	
2-Layer Sandwich	
3-Layer(aba) Sandwich	
3-Layer(bba) Sandwich	
3-layer(bab) Sandwich	

In CATH, the assignments of structures to fold groups and homologous superfamilies are made by sequence and structure comparisons.

Homologous domains, share common ancestor

Fold families

Class

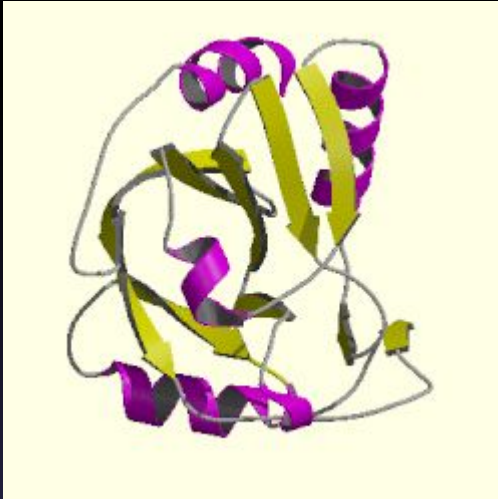
Architecture

Topology

H-level

Class 3

Alpha Beta



Architecture: 'Barrel'

D-amino Acid Aminotransferase; Chain A, domain 2	
L-fucose Isomerase; Chain A, domain 3	
Serine Protease, Human Cytomegalovirus Protease; Chain A	
Aconitase; domain 4	
TIM Barrel	

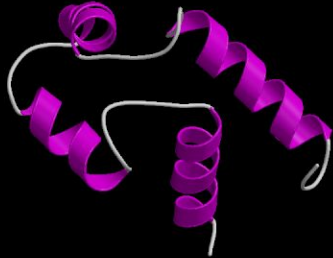
Homologous
domain family ?

9 Topologies : type of SS, number and order

Secondary structure prediction

- Historically first structure prediction methods predicted secondary structure
- Can be used to improve alignment accuracy
- Can be used to detect domain boundaries within proteins with remote sequence homology
- Often the first step towards 3D structure prediction
- Informative for mutagenesis studies

Protein Secondary Structures (Simplifications)



α -HELIX



β -STRAND



COIL (everything else)

Assumptions

- The entire information for forming secondary structure is contained in the primary sequence
- side groups of residues will determine structure
- examining windows of 13-17 residues is sufficient to predict secondary structure
 - α -helices 5–40 residues long
 - β -strands 5–10 residues long

Predicting Secondary Structure From Primary Structure

- accuracy 64-75%
- higher accuracy for α -helices than for β -sheets
- accuracy is dependent on protein family
- predictions of engineered (artificial) proteins are less accurate

A surprising result!



Chameleon
sequences

The “Chameleon” sequence

sequence 1

TEAVDA**AATAEKVFKQY**ANDNGVDG**EWTYDDATKKTFTVTEK**

sequence 2



Replace both sequences with
an engineered peptide (“chameleon”)

TEAVD**AWTVEKAFKTF**ANDNGVDG**AWTVEKAFKTF**TVTEK



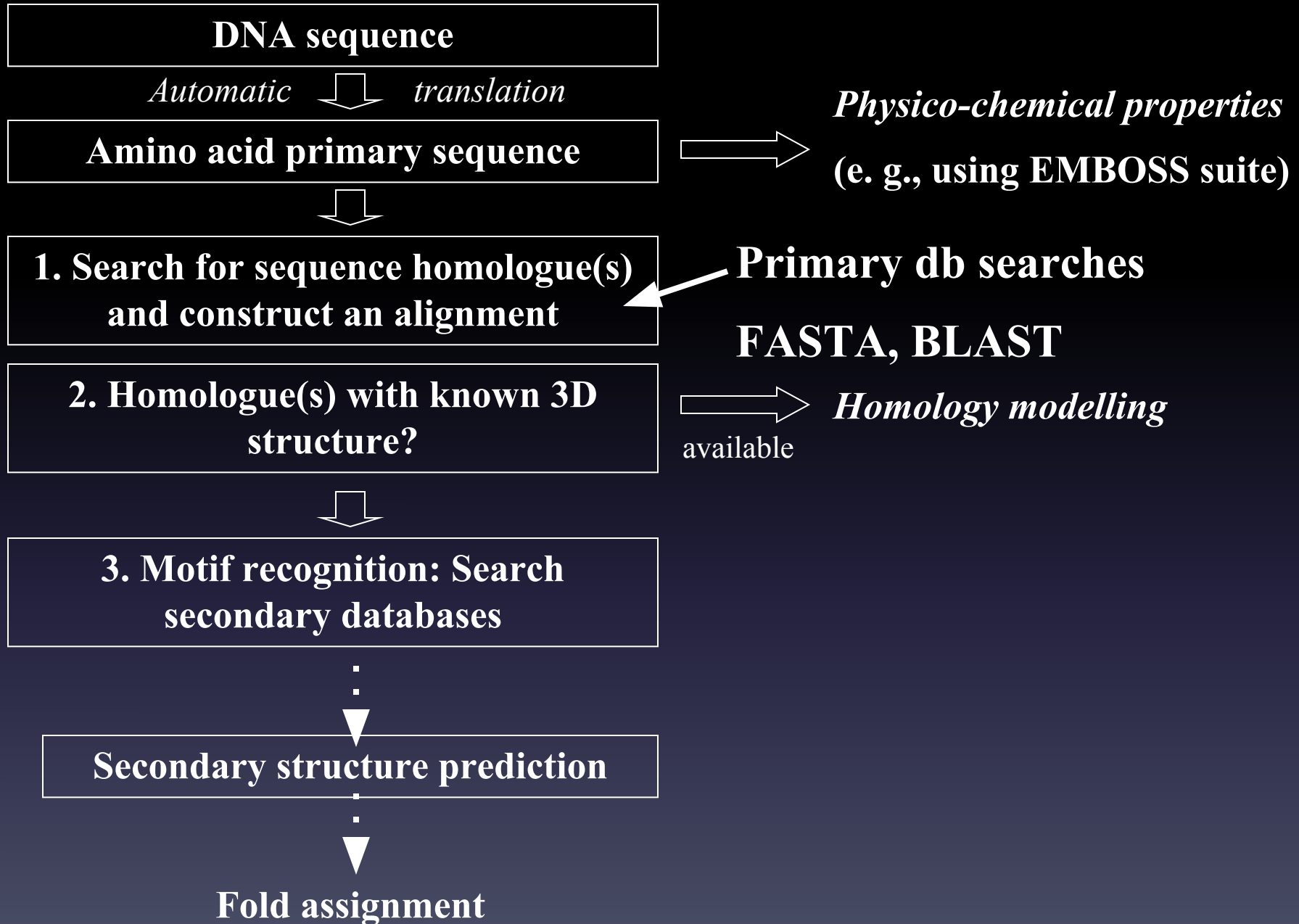
α -helix



β-strand

Source: Minor and Kim. 1996. *Nature* 380:730-734

<https://www.nature.com/articles/380730a0>



FASTA Format

- simple format used by almost all programs
- [**>**] header line with a [**hard return**] at end
- Sequence (no specific requirements for line length, characters, etc)

```
>U01 u01.seq Length: 2018 November 9, 2000 11:50 Type: N Check: 3854 ..
CGCAGAAAGAGGAGGCGCTTGCCTTCAGCTTGTGGGAAATCCCGAAGATGGCCAAAGACA
ACTCAACTGTTTCGTTGCTTCCAGGGCCTGCTGATTTTTTGGAAATGTGATTATTGGTTGTT
GCGGCATTGCCCTGACTGCGGAGTGCATCTTCTTTGTATCTGACCAACACAGCCTCTACC
CACTGCTTGAAGCCACCGACAACGATGACATCTATGGGGCTGCCTGGATCGGCATATTTG
TGGGCATCTGCCTCTTCTGCCTGTCTGTTCTAGGCATTGTAGGCATCATGAAGTCCAGCA
GGAAAATTCTTCTGGCGTATTTCAATTCTGATGTTTATAGTATATGCCTTTGAAGTGGCAT
CTTGTATCACAGCAGCAACACAACAAGACTTTTTTCACACCCAACCTCTTCCTGAAGCAGA
TGCTAGAGAGGTACCAAAACAACAGCCCTCCAAACAATGATGACCAGTGGAAAAACAATG
GAGTCACCAAAACCTGGGACAGGCTCATGCTCCAGGACAATTGCTGTGGCGTAAATGGTC
CATCAGACTGGCAAAAATACACATCTGCCTTCCGGACTGAGAATAATGATGCTGACTATC
CCTGGCCTCGTCAATGCTGTGTTATGAACAATCTTAAAGAACCTCTCAACCTGGAGGCTT
```

Examples

;LCBO - Prolactin precursor - Bovine

; a sample sequence in FASTA format

```
MDSKGSSQKGSRLLLLLLVVSNLLLCQGVVSTPVCPNGPGNCQVSLRDLFDRAVMVSHYIHD
LSS
EMFNEFDKRYAQGKGFITMALNSCHTSSLPTPEDKEQAQQTHHEVLMSLILGLLSWNDPLY
HL
VTEVRGMKGAPDAILSRAIEIEEENKRLLEGMEMIFGQVIPGAKETEPYPVWSGLPSLQTKDE
D
ARYSAFYNLLHCLRRDSSKIDTYLKLLNCRIIYNNNC*
```

>MCHU - Calmodulin - Human, rabbit, bovine, rat, and chicken

```
ADQLTEEQIAEFKEAFSLFDKDGDTITTKELGTVMRSLGQNPTEAELQDMINEVDADGNGTI
D
FPEFLTMMARKMKDSTDSEEEIREAFRVFDKDGNGYISAAELRHVMTNLGEKLTDEEVDDEMIR
EA
DIDGDGQVNYEEFVQMMTAK*
```

>gi|5524211|gb|AAD44166.1| cytochrome b [Elephas maximus maximus]

```
LCLYTHIGRNIYYGSYLYSETWNTGIMLLLITMATAFMGYVLPWQGMSFWGATVITNLFSAI
PYIGTNLV
EWIWGGGFSVDKATLNRFFAFHFILPFTMVALAGVHILTLHETGSNNPLGLTSDSDKIPFHPYY
TIKDFLG
```

- Anything other than a valid character would be ignored (including spaces, tabulators, asterisks, etc...).

- It was also common to end the sequence with an "*"

```
LLLLLLLLLLLLLSPDMLGDPDNHMPADPLNTPPHIKPEWYFLFAYAILRSVPNKLGGVLALF
LSIVIL
```


PDB format

index	resname		chain		resid	x	y	z	segname		
	name										
ATOM	22	N	ALA	B	3	-4.073	-7.587	-2.708	1.00	0.00	BH
ATOM	23	HN	ALA	B	3	-3.813	-6.675	-3.125	1.00	0.00	BH
ATOM	24	CA	ALA	B	3	-4.615	-7.557	-1.309	1.00	0.00	BH
ATOM	25	HA	ALA	B	3	-4.323	-8.453	-0.704	1.00	0.00	BH
ATOM	26	CB	ALA	B	3	-4.137	-6.277	-0.676	1.00	0.00	BH
ATOM	27	HB1	ALA	B	3	-3.128	-5.950	-0.907	1.00	0.00	BH
ATOM	28	HB2	ALA	B	3	-4.724	-5.439	-1.015	1.00	0.00	BH
ATOM	29	HB3	ALA	B	3	-4.360	-6.338	0.393	1.00	0.00	BH
ATOM	30	C	ALA	B	3	-6.187	-7.538	-1.357	1.00	0.00	BH
ATOM	31	O	ALA	B	3	-6.854	-6.553	-1.264	1.00	0.00	BH
ATOM	32	N	ALA	B	4	-6.697	-8.715	-1.643	1.00	0.00	BH
ATOM	33	HN	ALA	B	4	-6.023	-9.463	-1.751	1.00	0.00	BH
ATOM	34	CA	ALA	B	4	-8.105	-9.096	-1.934	1.00	0.00	BH
ATOM	35	HA	ALA	B	4	-8.287	-8.878	-3.003	1.00	0.00	BH
ATOM	36	CB	ALA	B	4	-8.214	-10.604	-1.704	1.00	0.00	BH
ATOM	37	HB1	ALA	B	4	-7.493	-11.205	-2.379	1.00	0.00	BH
ATOM	38	HB2	ALA	B	4	-8.016	-10.861	-0.665	1.00	0.00	BH
ATOM	39	HB3	ALA	B	4	-9.245	-10.914	-1.986	1.00	0.00	BH
ATOM	40	C	ALA	B	4	-9.226	-8.438	-1.091	1.00	0.00	BH
ATOM	41	O	ALA	B	4	-10.207	-7.958	-1.667	1.00	0.00	BH

To read up

- **Chou–Fasman method (Secondary structure)**
- **Smith–Waterman algorithm (Sequence)**
- **Needleman–Wunsch algorithm (Sequence)**